Journal of Hydrology 645 (2024) 132260

Contents lists available at ScienceDirect

Journal of Hydrology

journal homepage: www.elsevier.com/locate/jhydrol

Research papers

Response of urban flood resilience to climate change: An exploration with a novel performance-based metric considering the socioeconomic impacts of damage costs

Jiaxuan Zheng [a,b], Xi Chen [c], Kenji Kawaike [b], Kazuki Yamanoi [b], Takahiro Koshiba [b], Guoru Huang [a,d,e,*]

[a] School of Civil Engineering and Transportation, South China University of Technology, Guangzhou 510640, China
[b] Disaster Prevention Research Institute, Kyoto University, Kyoto 612-8235, Japan
[c] Department of Civil and Earth Resources Engineering, Kyoto University, Kyoto 612-8235, Japan
[d] State Key Laboratory of Subtropical Building and Urban Science, South China University of Technology, Guangzhou 510640, China
[e] Guangdong Engineering Technology Research Center of Safety and Greenization for Water Conservancy Project, Guangzhou 510640, China

ARTICLE INFO

Keywords:
Urban flood resilience
Socioeconomic impacts
Climate change
Flooding damage costs
Performance-based

ABSTRACT

Climate change stimulates the emergence of floods and increases flood risk for communities, resulting in extensive casualties and economic loss. Thus, flood resilience has been the focus of much attention. Integrated and holistic metrics are important for an accurate urban flood resilience evaluation. However, performance-based metrics involving surface inundation do not consider the socioeconomic impacts of damage costs on the recovery ability of the urban systems. This hinders the investigation of climate change-induced effects on flood resilience. Therefore, this study proposed a performance-based resilience metric considering the negative impacts of flooding damage costs. The proposed metric was then piloted in a highly urbanized area in Osaka City, Japan to explore the responses of flood resilience to climate change. The results suggest that climate change exerts non-negligible pressure on the study area to maintain its economic characteristics during flooding. Rainfall intensity primarily affects the urban system's most unfavorable state, while the total amount and duration of rainfall mostly impact the relatively stable state following flooding. Climate change severity is positively linked to the reduction in overall flood resilience. Continued climate change will further expand the spatial coverage of flood resilience losses on a global scale. Furthermore, the negative impacts of economic losses on flood resilience are more noticeable during rainfall events with higher rainfall intensities. This study improves the comprehensiveness of performance-based flood resilience evaluations and provides a reference for the effective enhancement of urban flood resilience under climate change.

1. Introduction

The frequency and severity of flooding have increased due to climate change, raising concerns about the harm it poses to populations (Tellman et al., 2021; Zou et al., 2013) and properties across the globe (Haer et al., 2020; Ma and Mostafavi, 2024). For example, it is documented that the global flooding disaster resulted in around 7 million deaths and damages of over US $600 billion from 1900 to 2013 (Chen et al. 2021a, b). Kumamoto Prefecture, Japan experienced massive flooding in July 2020, leaving 65 people dead and two missing and causing an extensive economic loss of 145.2 billion Yen (Department of Civil Engineering, 2020). The annual frequency of rainfall events with hourly intensity exceeding 50 mm/h in Japan increased by 1.4 times in the last 30 years due to climate change, while future rainfall events with short duration and high intensity and those with high rainfall volume are expected to occur more frequently (Ministry of Land, Infrastructure, Transport and Tourism, 2020). As a consequence, the probability of flood occurrence is expected to increase with more areas experiencing flooding.

In the light of the rising frequency and severity of global flooding issues, flood response to the changing climate has been hotly discussed (Fernández-Nóvoa et al., 2022; Padulano et al., 2021; Pal et al., 2023).

* Corresponding author at: School of Civil Engineering and Transportation, South China University of Technology, Guangzhou 510640, China.
E-mail address: huanggr@scut.edu.cn (G. Huang).

https://doi.org/10.1016/j.jhydrol.2024.132260

Available online 29 October 2024
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Urban flood resilience, one of the forward-looking core concepts in global resilient city construction (United Nations Office for Disaster Risk Reduction, 2019), has attracted widespread attention for disaster reduction and sustainable development. Holling (1973) first introduced "resilience" into the ecological domain and later reported the differences between ecological resilience and engineering resilience (Holling, 1996). Contemporary resilience research considers ecological resilience to emphasize a site's ability to adapt to external shocks and changes in controlling interactions, whilst engineering resilience highlights a site's ability to return to its pre-disturbance state (Pickett et al., 2014). Despite the early discussion on resilience, its application in disaster prevention such as flooding mitigation is a recent development and still needs in-depth exploration (Zheng and Huang, 2023a).

The failure of traditional flooding defense infrastructures such as levees, dams, and channelization to address the evolving flooding issues (Rözer et al., 2022) places a great emphasis on the need to develop urban flood resilience (Dahm, 2014). The concept of urban flood resilience supplements traditional flood risk management (Disse et al., 2020), while paving new ways for disaster reduction under climate change. Following the progress of research on resilience theme, the evaluation of resilience has increasingly moved from qualitative analysis (Kuang and Liao, 2020; Mavhura et al., 2013) to quantitative assessment (Haque et al., 2022; Moura Rezende et al., 2019). Performance-based approaches are one of the main quantitative assessment techniques for urban flood resilience. These approaches are centered on the system performance curves introduced by Bruneau et al. (2003), with early methods based on the structural performances of urban drainage systems (Dong et al., 2017; Mugume et al., 2015). However, as urban pluvial flooding is more closely associated with surface inundation (Chen et al., 2018), scholars have shifted their focus to surface inundation for proposing performance-based urban flood resilience evaluation methods. These metrics are typically employed with urban flood models providing time-dependent variables such as surface water depth and velocity, thus reflecting the dynamic flooding impacts on normal operations of urban systems during a specific event (Chen et al. 2021a, b; Leandro et al., 2020; Wang et al., 2019; Zheng and Huang, 2023b).

Despite the great efforts in urban flood resilience assessments, the majority of the existing metrics based on system performance curve and surface inundation depend on the flooding impacts on the physical environment, namely, the impacts on the functionality and characteristics of different land use types (Chen et al. 2021a, b; Wang et al., 2019). Although flooding impacts on social components have also been considered in the metric proposed by Zheng and Huang (2023b), they only involve the stability of residents and vehicles on roads, neglecting the flooding economic loss. This prevents the socioeconomic impacts of flooding from being comprehensively reflected in flood resilience evaluations based on system performance curves. In addition, flooding damage costs should be accounted for flood resilience evaluations as they are closely related to the post-disaster recovery of urban systems (Ballesteros et al., 2023). Since the economy is an important dimension of urban disaster resilience (Büyüközkan et al., 2022; Tong, 2021), the consideration of economy-related factors yields benefits for the comprehensiveness of the performance-based resilience metric (Prashar et al., 2023). Furthermore, given the growing global concerns about flooding, it is crucial to determine how flood resilience responds to climate change using a comprehensive and integrated flood resilience assessment approach.

This study adopts the widely used definition of urban flood resilience from the existing performance-based resilience research, namely the system's capacity to return to its pre-disrupted state following a flooding disturbance (Chen et al. 2021a, b; Leandro et al., 2020). This definition can be categorized as engineering resilience (Chen et al. 2021a, b). Given the aforementioned knowledge gaps and concerns, this study aims to: (1) propose a novel performance-based urban flood resilience metric considering the socioeconomic impacts of flooding damage costs; (2) extract representative rainfall events under different climate scenarios; (3) calculate urban flood resilience under different climate scenarios based on a one dimensional (1D)–two-dimensional (2D) hydrodynamic flood model; and (4) explore the impacts of climate change on urban flood resilience. The remainder of this paper is structured as follows. Section 2 details the materials and methodological approach used in this study. Section 3 describes the flooding damage costs factor, system performance, and urban flood resilience results under different scenarios. Section 4 discusses the results and presents an outlook for further studies. Section 5 summarizes the key findings of the study.

2. Materials and methods

2.1. Study area

The study area is highly urbanized (Fig. 1) and densely populated and is located in the Nakahama area in Osaka City, Japan. Covering an area of 18 km^2, it has multiple major railway stations including Tennoji, Kyobashi, and Tsuruhashi, and numerous railway lines. The terrain is high in the west, with a gentle north–south slope, and three sub-regions (A, B, and C in Fig. 1) divided by waterbodies. The levees along rivers can prevent the study area from suffering fluvial flooding. However, the low topography and increased difficulty in flow draining from its enclosure by levees result in pluvial flooding caused by heavy rainfall, such as the flooding on August 27, 2011 (Lee et al., 2016) and August 25, 2013 (Kawaike and Nakagawa, 2019).

2.2. Flood model establishment

The flood model established in this study is based on the 1D-2D hydrodynamic flood model. The 1D-2D hydrodynamic flood model was first developed by Lee et al. (2016) and later improved by Sawada (2016), combining a 1D sewerage-system model and a 2D ground-surface model with an interaction model. The applied DEM has a spatial resolution of 5 m and can be downloaded from the website https://fgd.gsi.go.jp/download/menu.php. The land use distribution in the model is determined following Fig. 1. The sewage system in the study area was generalized into 4,317 manholes, 4,445 pipes, and five pump stations in the built flood model. Kawaike and Nakagawa (2019) present the sewerage system structure. A total of 109,034 unstructured triangular meshes were generated in the 2D model considering building and road network configurations, with a maximum area of 922.85 m^2 and a minimum area of 12.46 m^2. The elevation of meshes in roads is lowered by 20 cm to represent the flooding discharge effects (Huang et al., 2019). The built model of the study area was validated using a rainfall event on August 25, 2013. The comparison between the simulated and observed data indicates that the model can be applied to urban pluvial inundation simulation (Kawaike and Nakagawa, 2019). Saito et al. (2021) also adopted the developed model for urban inundation analysis.

2.3. SI-CAT climate dataset

We employed the climate dataset from the Social Implementation Program on Climate Change Adaption Technology (SI-CAT) (https://doi.org/10.20783/DIAS.562). This 5-km dataset, which spans Honshu, Shikoku, and Kyushu islands, was dynamically downscaled from a 20-km large-ensemble climate dataset from the Database for Policy Decision-Making for Future Climate Changed (d4PDF) (Ishii and Mori, 2020) using the Non-hydrostatic Regional Climate Model (NHRCM) (Sasaki et al., 2008). The model settings for dynamic downscaling were validated using the Japanese 55-year reanalysis data set (JRA55) (Kobayashi et al., 2015) and the evaluation results exhibit a good agreement with the observation (Kawase et al., 2018; Sasai et al., 2019). In addition to the climate model of GFDL-CM3 (Griffies et al., 2011) selected in this study, five climate models, namely CCSM4 (Gent et al., 2011); HadGEM2-AO (Baek et al., 2013); MIROC5 (Watanabe

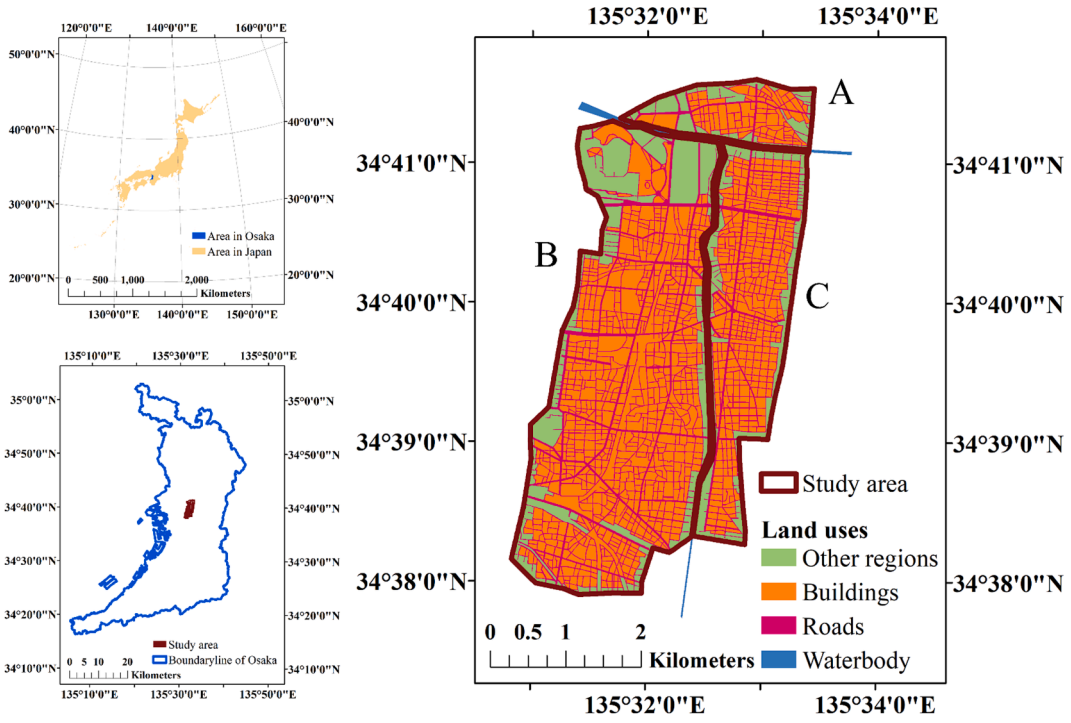


Fig.1. The illustrative map of the study area.

et al., 2010); MPI-ESM-MR (Shi et al., 2018); and MRI-CGCM3 (Zhang et al., 2019) are available in the dataset. Three climate scenarios are provided for each climate model. Besides the present climate scenario, future scenarios assuming a global mean temperature increase by 2 K (2℃) in the near future and 4 K (4℃) in the far future from the preindustrial period are also included. The simulations of two future warming climate scenarios are based on the Representative Concentration Pathway of 8.5 scenarios in 2040 for the 2 K scenario and 2090 for the 4 K scenario. For example, the boundary condition of greenhouse gases in the weather prediction model was set as values at 2040 and 2090 of RCP 8.5 for the 2 K and 4 K scenarios, respectively. Further details can be referenced to Fujita et al. (2019) and Mizuta et al. (2017).

Multiple experiments with different lateral boundary conditions and perturbations were performed and termed as different members in the dataset (Sasai et al., 2019). Further details can be found at https://www.miroc-gcm.jp/d4PDF/design.html. In SI-CAT, 12 members are included in each climate scenario, with each member covering 31 years.

2.4. Rainfall events

Hourly rainfall data was derived using the Grid Analysis and Display System (GrADS) based on the longitude and latitude of the study area. We adopted rainfall data from the GFDL-CM3 climate model for the present and future experiments provided in the SI-CAT dataset, with the period from 1980 to 2010 employed for the present experiment and 2060 (2080) to 2090 (2110) for the future scenario of 2 K (4 K) warmer than the preindustrial temperature. Specifically, climate data from August 16, 1980 to August 15, 2010 under the present scenario; August 16, 2060 to August 15, 2090 under the 2 K scenario; and August 16, 2080 to August 15, 2110 under the 4 K scenario were extracted following the relative document (SI-CAT DDS5TK team, 2022).

The rainfall data of each climate scenario was divided into independent events if the intensity was less than 0.5 mm/h for more than 1 h continuously based on previous work (Huang et al., 2021; Song et al., 2015) and the rainfall data recorded by the Japan Meteorological Agency. The total rainfall for each event was arranged in descending order under each climate scenario to determine the events with maximum rainfall. When comparing the maximum total rainfall under different scenarios, the one under the 4 K scenario is the highest, followed by that under the 2 K scenario, and the one under the present scenario is the lowest. Since the rainfall data we used for all scenarios spans 30 years, the selected three rainfall events have the same probability of occurrence under their respective scenarios (1/30). Therefore, under the premise of the same probability of occurrence, the three rainfall events are consistent with the objective cognition that the higher the temperature is, the higher the total rainfall is, which indicates the reliability of the division method. Fig. 2 depicts the hyetographs of the selected rainfall events and Table 1 provides relevant information. Although the rainfall event of the 4 K future scenario has the highest total rainfall and longest duration, its peak rainfall intensity is lower than that of the 2 K future scenario. The total rainfall and peak rainfall intensity of the events during the future period are higher than those under the present scenario, yet the event of the 2 K future scenario has the shortest duration.

2.5. Flooding damage costs calculation

The flooding damage costs were mesh-based and determined by following official documents (Water and Disaster Management Bureau, 2023, 2020) and previous research (Akiyama et al., 2012; Kawaike and Nakagawa, 2018; Ko et al., 2018). Meshes were from the flood model described in Section 2.2, whose land uses were classified based on Fig. 1. The costs included those from residential houses, household assets, and office properties (Kawaike and Nakagawa, 2018; Water and Disaster Management Bureau, 2020). Fig. 3 presents the distribution of the aforementioned building types in the study area, which serves as the basis for identifying the building types of meshes with buildings.

2.5.1. Damage costs of residential houses

The damage costs of residential houses in mesh i (C_{rh}) are governed by Eq. (1).

$$C_{rh}(i) = A_{rh}(i) \cdot V_{rh} \cdot DR(i) \quad (1)$$

where A_{rh} (i) is the area of residential houses in mesh i (m^2); V_{rh} is the

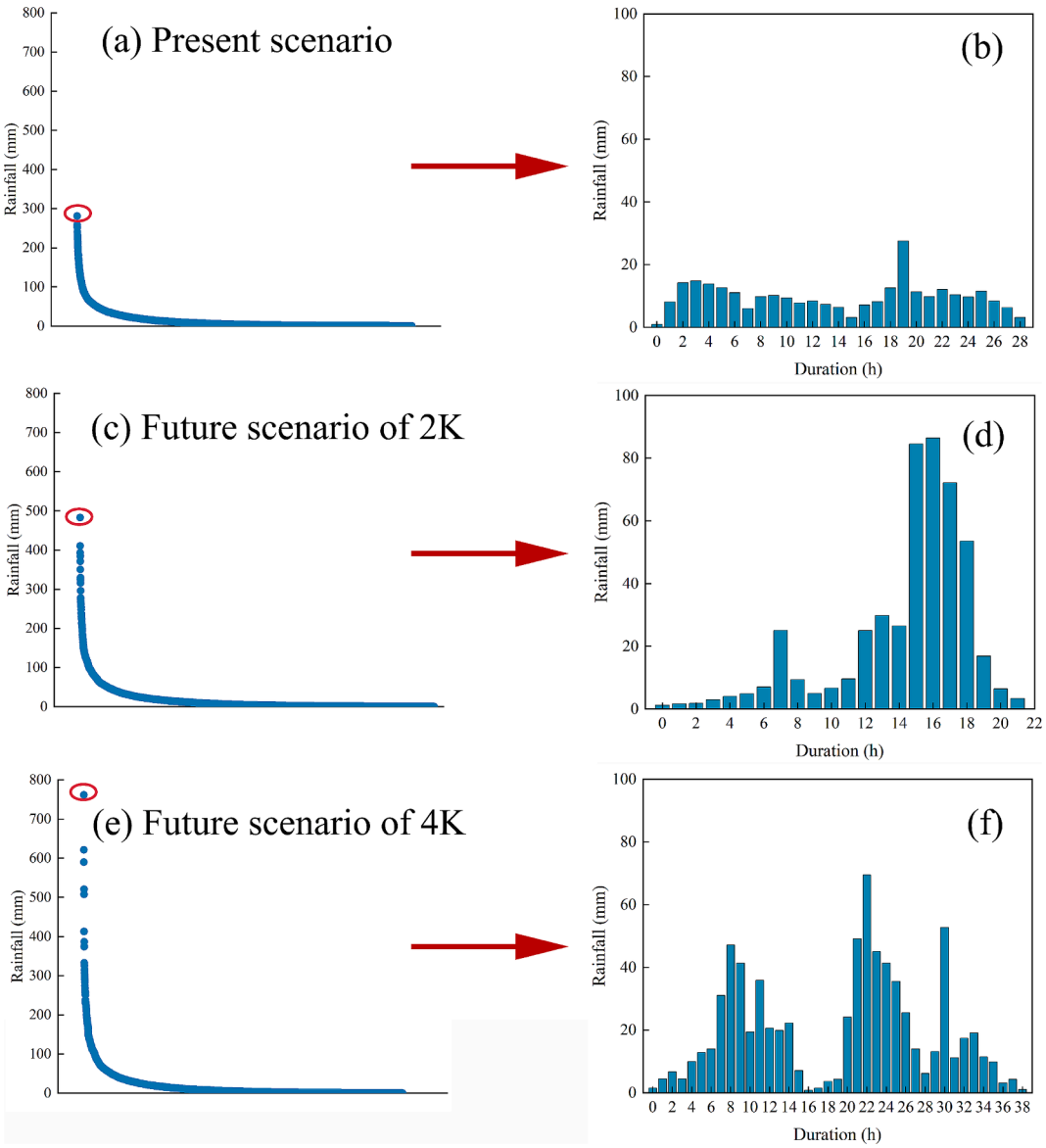


Fig.2. Rainfall events under different climate scenarios.

Table 1
Summary of rainfall events under different climate scenarios.

Scenarios	Total rainfall (mm)	Peak intensity (mm/h)	Duration (h)
Present scenario	281.07	27.45	28
Future scenario under 2 K	483.10	86.33	21
Future scenario under 4 K	761.42	69.43	38

unit value of residential houses (Yen/m^2); and $DR(i)$ is the damage rate of mesh i, which is related to the ground gradient and the maximum surface inundation depth during a specific flooding event (Water and Disaster Management Bureau, 2020).

Based on the latest data (Water and Disaster Management Bureau, 2023), V_{rh} in Osaka City is determined as 229,100 Yen/m^2. Table 2 reports the values of DR, considering that the ground gradients of the residential houses in the study area are less than 1/1,000 (Ko et al., 2018) and the damage rate equals 0 if the maximum inundation depth is less than 0.2 m (Kawaike and Nakagawa, 2018).

2.5.2. Damage costs of household assets

The damage costs of household assets in mesh i (C_{ha}) are calculated as follows:

$$C_{ha}(i) = N(i)\cdot\alpha(i)\cdot V_{ha}\cdot DR(i) \quad (2)$$

where $N(i)$ is the number of households in mesh i, and every household is assumed to have a residential area of 70 m^2 (Kawaike and Nakagawa, 2018); $\alpha(i)$ is the correction coefficient of the residential region, which equals 0.5 for the detached houses and 0.2 for the housing complexes (Kawaike and Nakagawa, 2018); V_{ha} is the appraised assets value per household, including automobiles (3,169 Yen per household) and other types of household assets (9,368 Yen per household) (Water and Disaster Management Bureau, 2023). Tables 2 and 3 report the values of DR for household assets excluding automobiles and automobiles, respectively.

2.5.3. Damage costs of office properties

Office properties are classified into two types, namely, depreciable assets and inventory assets, with corresponding damage costs quantified as follows:

$$C_{da}(i) = N_e(i)\cdot\beta\cdot V_{da}\cdot DR(i) \quad (3)$$

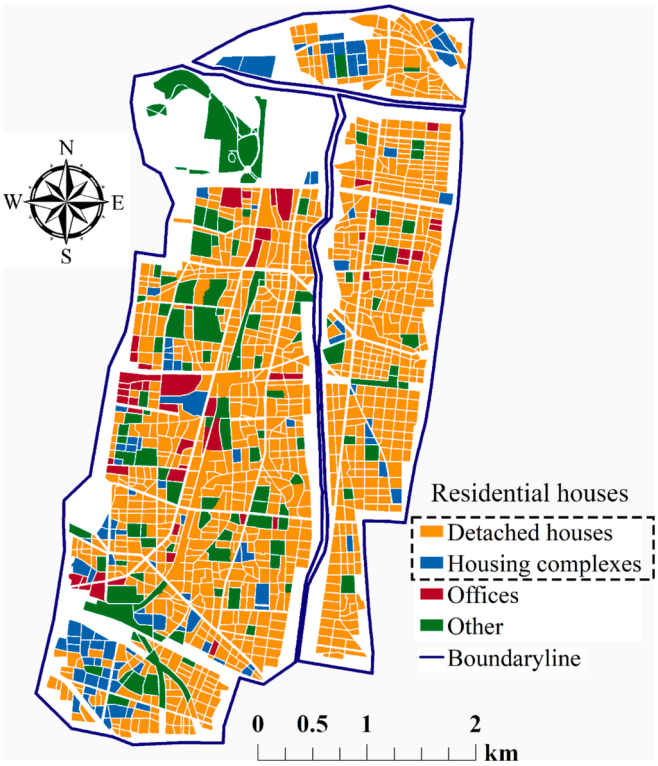


Fig.3. Spatial pattern of different building types.

Table 2
Values of DR of various asset types and maximum water depth (Water and Disaster Management Bureau, 2020).

Maximum water depths h (m)	Asset types			
	Residential houses	Household assets excluding automobiles	Office depreciable assets	Office inventory assets
$h < 0.2\ m$	0	0	0	0
$0.2\ \text{m} \le h < 0.45\ m$	0.047	0.037	0.064	0.053
$0.45\ \text{m} \le h < 0.5\ m$	0.189	0.308	0.296	0.282
$0.5\ \text{m} \le h < 1\ m$	0.253	0.533	0.573	0.440
$1\ \text{m} \le h < 2\ m$	0.406	0.701	0.801	0.814
$2\ \text{m} \le h < 3\ m$	0.592	0.948	0.920	0.946
$h \ge 3\ m$	0.800	0.977	0.940	0.975

Table 3
Values of DR of automobiles under different water depths (Water and Disaster Management Bureau, 2020).

Maximum water depths h (m)	DRs
$0 < h < 0.3\ m$	0
$0.3\ \text{m} \le h < 0.5\ m$	0.1
$0.5\ \text{m} < h < 0.7\ m$	0.5
$h \ge 0.7\ m$	1

$$C_{ia}(i) = N_e(i) \cdot \beta \cdot V_{ia} \cdot DR(i) \tag{4}$$

where $C_{da}(i)$ and $C_{ia}(i)$ are the damage costs of office depreciable assets and inventory assets in mesh i, respectively; $N_e(i)$ is the number of office employees in mesh i; β is the correction coefficient of the office area, which equals 0.286 in this study (Kawaike and Nakagawa, 2018); and V_{da} and V_{ia} are the appraised values of depreciable and inventory assets per office employee, respectively. $DR(i)$ is determined based on the data presented in Table 2.

There are 2,394,461 office employees in Osaka City (Osaka Prefectural Government, 2023) and the city covers an area of 225 km^2 (https://www.city.osaka.lg.jp/toshikeikaku/page/0000402930.html). Thus, the number of office employees in the study area can be estimated as 191,557 based on the area proportion of the study area to Osaka City. $N_e(i)$ is related to the area proportion of mesh i in the office area and the number of employed officers in the study area (Kawaike and Nakagawa, 2018). Based on the appraised values listed in the manual (Water and Disaster Management Bureau, 2023), V_{da} and V_{ia} are determined as 8,569,000 Yen and 4,490,000 Yen, respectively.

2.6. Urban flood resilience calculation

The mesh-based metric without flooding damage costs proposed by Zheng and Huang (2023b) was used as the basis of the urban flood resilience metric we proposed in this study. The metric reveals how the development of surface flooding acts on the functionalities and characteristics of an urban system, which can be depicted as a system performance curve (Fig. 4). The duration of system performance being impacted (DSPI) and the flooding impacts on the stability of residents and vehicles on the traffic land are considered in the urban flood resilience metric. In Fig. 4, the dashed line ($p_0 = 1$) represents the initial and highest performance of a mesh. The area enclosed by the solid and dashed lines denotes the flood severity. t_s is the moment when the system performance begins to decrease; t_{fs} is the moment when the system performance reaches the lowest value of p_f; t_{fe} is the moment when the system performance begins to recover from the most unfavorable flooding impacts; t_e is the moment when the system performance reverts to its initial state; t_n is the simulation duration; and t_s to t_e is the DSPI. Land use types of the study area are categorized into non-traffic and traffic land due to different calculation methods for system performance. The system performance of each mesh in non-traffic land is determined as follows:

$$p_{\text{non - traffic land}}(i,t) = \begin{cases} 1, & HR(i,t) \in [0, 0.75) \\ \dfrac{2.5 - HR(i,t)}{1.75}, & HR(i,t) \in [0.75, 2.5) \\ 0, & HR(i,t) \in [2.5, +\infty) \end{cases} \tag{5}$$

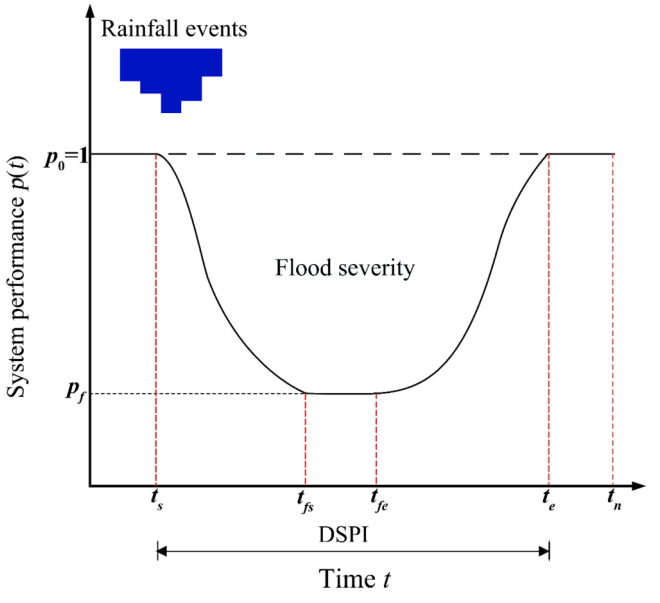


Fig.4. The illustration of a system performance curve.

where $p_{\text{non - traffic land}}(i,t)$ is the system performance of mesh i in non-traffic land at time t; and $HR(i,t)$ is the Flood Hazard rating index (HR Wallingford et al., 2006) of mesh i at time t, revealing the flood hazard faced by individuals and can be established by Eq. (6):

$$HR(i,t) = d(i,t)\cdot(v(i,t)+0.5)+DF(i) \tag{6}$$

where $d(i,t)$ and $v(i,t)$ are the water depth in mesh i at time t (m) and the velocity in mesh i at time t (m/s), respectively; and $DF(i)$ is the debris factor of mesh i, corresponding to the debris carried by floods. Table 4 reports the values of DF under different water depths, velocities, and land use types.

Considering the physical mechanism of residents and vehicles maintaining their stability during floods (Wang et al., 2021; Xia et al., 2016, 2014), the system performance of meshes in traffic land is expressed as in Eq. (7):

$$p_{\text{traffic land}}(i,t) = \begin{cases} 1 - \dfrac{v(i,t)}{U_c(i,t)}, v(i,t) < U_c(i,t) \\ 0, v(i,t) \geqslant U_c(i,t) \end{cases} \tag{7}$$

where $p_{\text{traffic land}}(i,t)$ is the system performance of mesh i in traffic land at time t; and $U_c(i,t)$ is the incipient velocity of mesh i at time t (m/s). This variable denotes the critical water velocity value that causes instability of residents or vehicles and is calculated using Eq (8). Note that when $d(i,t)$ surpasses 1.52 m, $p_{\text{traffic land}}(i,t)$ becomes 0, irrespective of the relationship between $v(i,t)$ and $U_c(i,t)$. Further details on the metric can be found in Zheng and Huang (2023b).

$$U_c(i,t) = \begin{cases} 9.684\left(\dfrac{d(i,t)}{1.26}\right)^{0.109}\sqrt{\dfrac{0.0202}{d(i,t)} - (0.007d(i,t)+0.005)}, d(i,t) \in (0, 0.16] \\ 6.034\left(\dfrac{d(i,t)}{1.26}\right)^{0.383}\sqrt{\dfrac{0.0255}{d^2(i,t)} - \left(0.01 + \dfrac{0.004}{d(i,t)}\right)}, d(i,t) \in (0.16, 0.3311] \cup (0.3881, 0.553] \cup [0.6, 1.01] \\ 0.492\left(\dfrac{d(i,t)}{1.48}\right)^{-0.344}\sqrt{\dfrac{9.122}{d(i,t)} - 23.505}, d(i,t) \in (0.3311, 0.3881] \\ 0.367\left(\dfrac{d(i,t)}{1.737}\right)^{-0.451}\sqrt{\dfrac{13.705}{d(i,t)} - 21.416}, d(i,t) \in (0.553, 0.6] \\ 6.034\left(\dfrac{d(i,t)}{1.77}\right)^{0.383}\sqrt{\dfrac{0.0751}{d^2(i,t)} - \left(0.017 + \dfrac{0.011}{d(i,t)}\right)}, d(i,t) \in (1.01, 1.52] \end{cases} \tag{8}$$

The higher the flooding damage costs, the more difficult and time-consuming it is for the community to recover to its pre-disrupted economic state. Based on this, a novel flood resilience metric was proposed by integrating the damage costs-related index into the metric proposed by Zheng and Huang (2023b) and is expressed as follows:

$$R(i) = \alpha(i)\cdot R_0(i) \tag{9}$$

Table 4
Values of DF under different conditions.

Water depths d (m) or velocities v (m/s)	Land use types		
	Pasture/Arable	Woodland	Urban
$0 \leq d < 0.25\ m$	0	0	0
$0.25\ \text{m} \leq d < 0.75\ m$	0	0.5	1
$d \geq 0.75$ m and/or $v \geq 2$ m/s	0.5	1	1

where $R(i)$ is the flood resilience of mesh i; $\alpha(i)$ is the flooding damage costs factor (Eq. (10)), whose calculation was determined by referencing relevant studies (Li et al., 2017; Zheng and Huang, 2023a); and $R_0(i)$ is the flood resilience ignoring flooding damage costs, which can be described by Eq. (11) (Zheng and Huang, 2023b):

$$\alpha(i) = 0.95\frac{C_{\max} - C(i)}{C_{\max} - C_{\min}} + 0.05 \tag{10}$$

where $C(i)$ is flooding damage costs in mesh i (Yen); and $C_{\max}$ and $C_{\min}$ are the maximum and minimum flooding damage costs in the study area, respectively (Yen).

$$R_0(i) = \left(\frac{1}{t_n}\int_0^{t_n} p(i,t)\mathrm{d}t\right)^{[DSPI(i)/t_n]} \tag{11}$$

where t_n is the simulation duration in the model (min), which is determined by the rainfall duration, the moment when surface water depth and velocity reach their stability, and the calculation complexity of quantifying resilience; and $DSPI(i)$ is the DSPI of mesh i (min).

Note that flooding damage costs only occur in areas with residential houses and offices (Section 2.5) based on the official documents and previous research (Akiyama et al., 2012). Flood resilience in other land use types is calculated following the existing method (Zheng and Huang, 2023b), with $\alpha(i)$ set as 1.

The system performance at various moments and urban flood resilience of a region comprised of multiple meshes ($p_{\text{overall}}(t)$ and R_{overall}) are equal to the average of the system performance and resilience of all meshes weighted by their area, as in Eqs. (12) and (13):

$$p_{\text{overall}}(t) = \sum_{i=1}^{j}\frac{A(i)}{A_{\text{overall}}}\cdot p(i,t) \tag{12}$$

$$R_{\text{overall}} = \sum_{i=1}^{j}\frac{A(i)}{A_{\text{overall}}}\cdot R(i) \tag{13}$$

where j is the number of total meshes within the selected region; A_{overall} is the total area of the selected region (m^2); and $A(i)$ is the area of mesh i in the selected region (m^2).

The resilience area of a mesh is defined as the product of the total area of the mesh and its flood resilience (Chen et al. 2021a, b). The urban flood resilience area loss (RAL) can be further determined using Eq. (14), which considers both the degree and extent of the resilience loss in a mesh and reveals the actual extent of the area where resilience is affected. RAL of different regions (RAL_{overall}) can be calculated following Eq. (15).

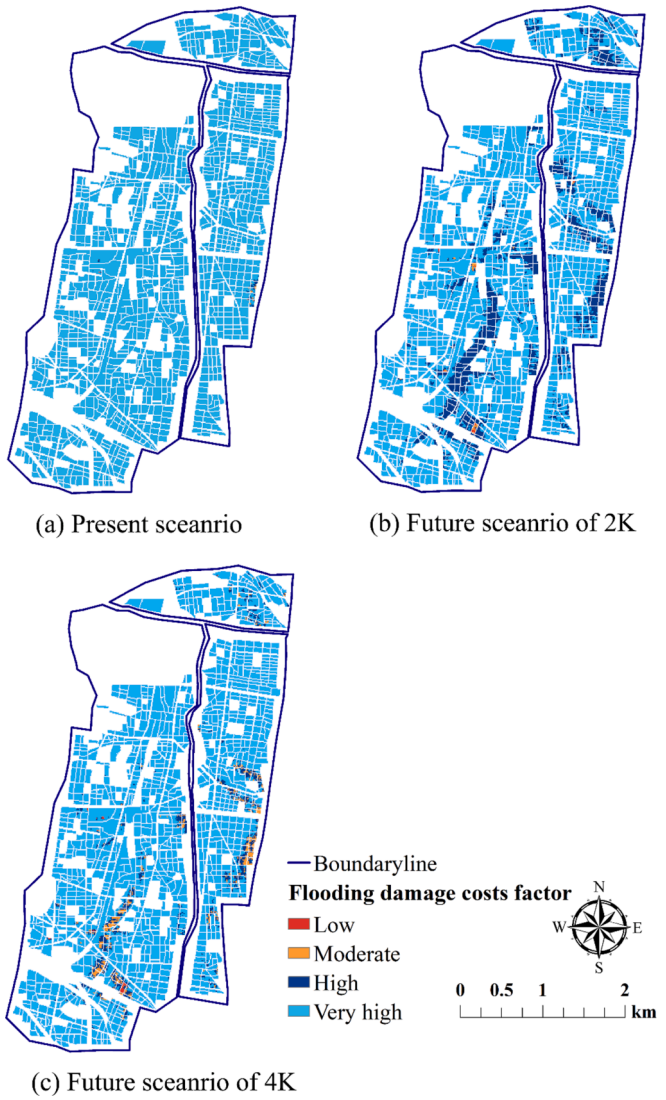


Fig.5. Flooding damage costs factor of specific building categories under different scenarios. The lower the flooding damage costs factor, the more severe the direct economic losses in this area compared to other areas within the study area.

$$RAL(i) = A(i)\cdot(1 - R(i)) \tag{14}$$

$$RAL_{\text{overall}} = \sum_{i=1}^{j} RAL(i) \tag{15}$$

3. Results

3.1. Flooding damage costs factor

The lower the flooding damage costs factor, the more severe the direct economic losses in this area compared to other areas within the study area (Section 2.6). Referencing Section 2.6, the flooding damage costs factor of land use types excluding those in Section 2.5 will remain unchanged ($\alpha = 1$). Thus, this section focuses on the flooding damage costs factor of the building categories described in Section 2.5.

Fig. 5 presents the spatial patterns of the flooding damage costs factor, with the ordinal scale classified by the Natural Breaks (Jenks) method under different climate scenarios. The vast majority of buildings have similar flooding damage costs under the present scenario, with a minimal proportion of areas experiencing typically high flood damage. However, this trend changes with the inclusion of climate change. Although most regions with buildings still do not suffer much economic loss (Very high level of flooding damage costs factor), marked expansions of areas with relatively higher economic damage (Low, Moderate, and High levels of flooding damage costs factor) are observed under future scenarios, particularly under the 2 K scenario. This may be explained by the more widely distributed flooded areas caused by the highest peak rainfall intensity occurring in the 2 K scenario compared to the other scenarios. The 4 K scenario exhibits the greatest distribution of areas with low and moderate levels of flooding damage costs factor. This suggests that the number of areas experiencing more severe economic losses increases with the temperature changes. Climate change exerts non-negligible pressure on the study area to maintain its economic characteristics during flooding.

3.2. System performance

Fig. 6 depicts the overall system performance of the study area and the three sub-regions in Fig. 1 at different times. All regions operate smoothest under the present scenario, with the lowest system performance values observed when the surface inundation reaches its relative stability under the 4 K future scenario. In addition, during the same period, regions finally transition from a more unfavorable condition to a more favorable state under the 2 K scenario compared to the 4 K scenario. This indicates the weakest recovery and adaptive capacity of different region types when the temperature is maximized. For the lowest system performance values under each scenario, all regions exhibit the lowest values under the 2 K future scenario. This reveals that the operational state of each region under the 2 K scenario differs most from its non-flooded state when it is experiencing the most serious damage to system functionality. Furthermore, given the differences between rainfall events under different scenarios, climate change exerts a significantly negative impact on the current state of the system in a variety of regions. In particular, rainfall intensity primarily affects the system's most unfavorable state, while the total amount and duration of rainfall mostly impacts the system's relatively stable state following flooding.

The moment when the system performance reaches its lowest occurs later than the moment when rainfall intensity reaches its peak under all scenarios. This indicates that the function loss in an area is consistently maximized after the rain peak occurs (Chen et al. 2021a, b). This may be due to the time lags between the most serious surface inundation and peak rainfall (Chen et al., 2018). The system performance in each region type exhibits more instability under the 4 K scenario. Further comparing Figs. 2 and 6 reveals similar trends between the system performance curves and rainfall event patterns.

3.3. Urban flood resilience

Table 5 reports the area proportions of urban flood resilience under each scenario, with the five levels determined by the Natural Breaks (Jenks) method. The urban flood resilience of most areas consistently remains at the highest level. This agrees with the results obtained in other studies (Chen et al. 2021a, b; Zheng and Huang, 2023b), implying that most areas can well maintain their initial functionality even under intense rainfall. Cities are able to endure adverse flooding impacts, which allows the urban system to retain its initial operation and characteristics within specific water depth and velocity ranges. Therefore, most areas are likely to remain highly resilient to flooding under climate change, despite the increasingly severe flooding in the study area (Saito et al., 2021). Regions with a very low flood resilience compromise the lowest proportion under all scenarios. Very low, low, medium, and high flood resilience areas expand their distribution under climate change, while very high flood resilience follows the opposite trend. This suggests that climate change contributes to a shift in flood resilience from the highest class to the lower classes, and such shifts are more likely to occur

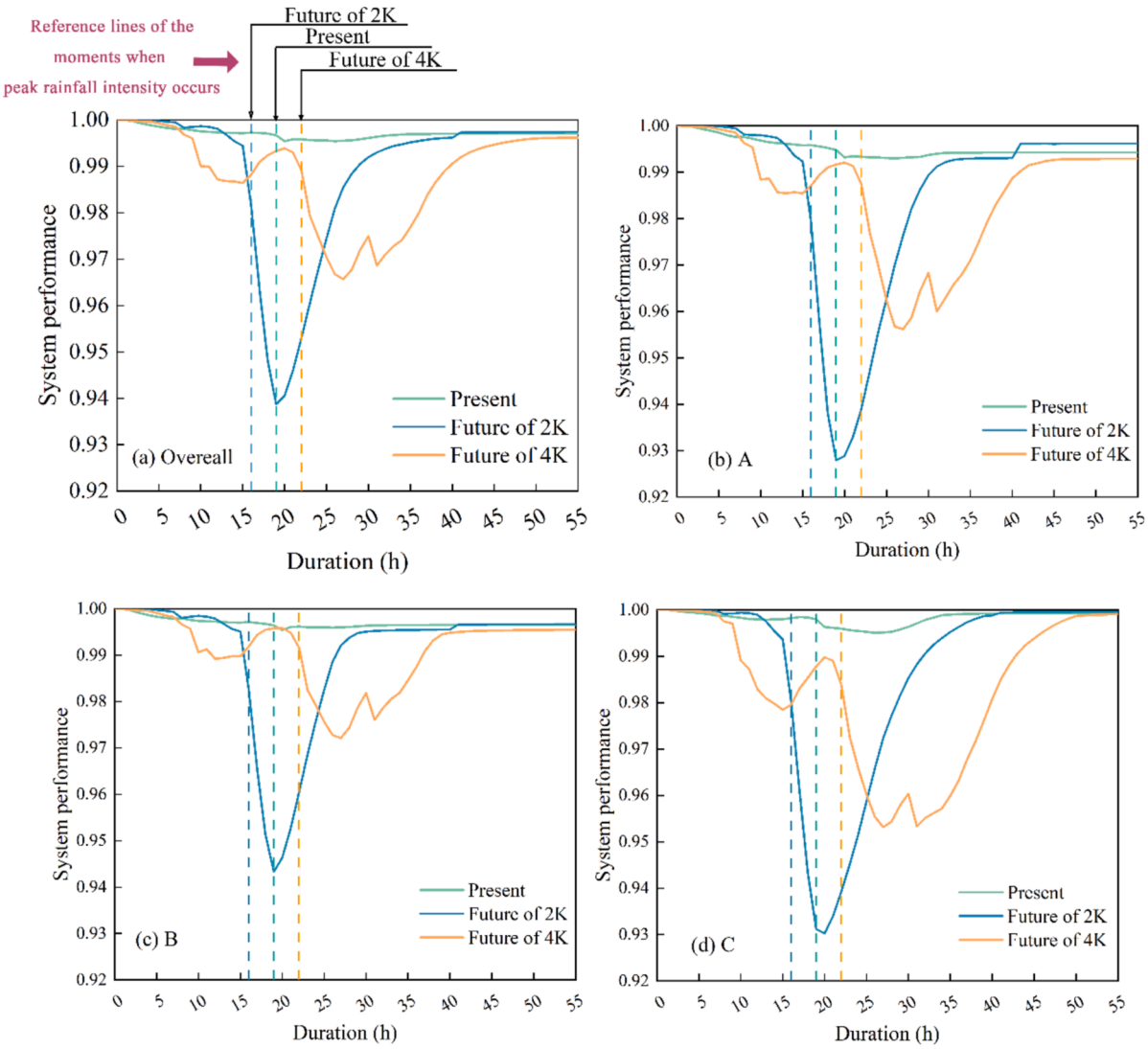


Fig.6. System performance of different regions by time under present and future scenarios.

Table 5
Area proportions of urban flood resilience of different levels under different scenarios (Unit: %).

Levels	Scenarios		
	Present scenario	Future scenario of 2 K	Future scenario of 4 K
Very low	0.01	0.31	0.16
Low	0.13	2.85	2.29
Moderate	0.21	4.93	2.21
High	1.41	10.60	6.80
Very high	98.25	81.31	88.53

between high and very high classes (Table 5). For the future scenarios, despite the lower total rainfall and shorter duration of the rainfall event under the 2 K scenario compared to that under the 4 K scenario, the coverage of flood resilience in the highest class under the 2 K scenario contracted compared to the 4 K scenario, while the opposite is true for other classes. This implies that the high peak rainfall intensity promotes the expansion of low-flood resilience areas.

Fig. 7 depicts the overall urban flood resilience of the study area and three sub-regions (Fig. 1). All regions exhibit high flood resilience close to 1 under the present scenario. This indicates that pluvial flooding under the present scenario does not have a significant impact on the original functioning and maintenance of the areas. However, this is not the case under future scenarios. For the overall study area, urban flood resilience gradually decreases as temperature rises, highlighting the positive relationship between climate change severity and the reduction in overall flood resilience. This may be due to the dramatic reduction in flood resilience in Region C with increasing temperature, while Regions A and B exhibit a slight increase in flood resilience in the 4 K scenario compared to the 2 K scenario. Further considering that the topography in the study area is high in the west and low in the east, Region C, which is situated in the eastern portion, is more susceptible to pluvial flooding. This is particularly true given that the study area will experience more severe flooding as a result of climate change. Hence, the functionality maintenance and normal operation of Region C should be a focus under climate change.

Although areas with low flood resilience are more numerous under the 2 K scenario compared to the 4 K scenario, the original characteristics of the study area are impaired more under the latter. This suggests that the functionality impairment of areas where flood resilience is impacted is more serious as climate change persists.

4. Discussion

4.1. Qualitative analysis of the proposed flood resilience metric

Conducting qualitative analysis can aid in ensuring that the proposed

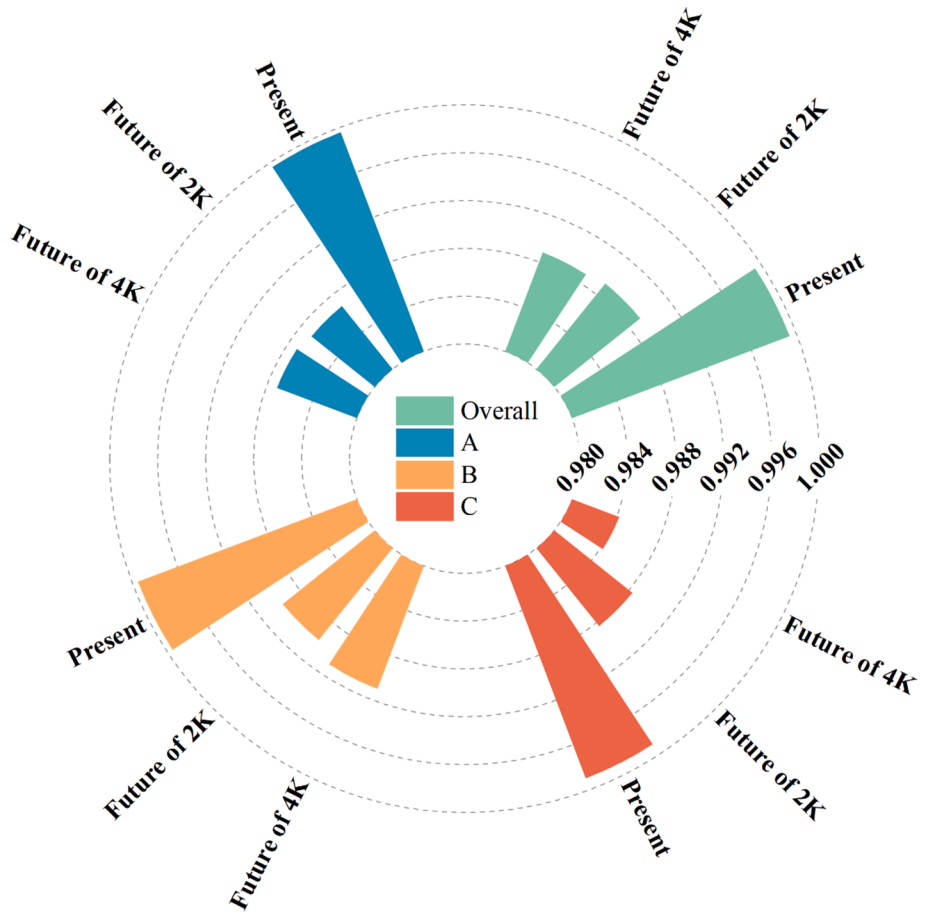


Fig.7. Urban flood resilience of different regions under present and future scenarios.

flood resilience metric conforms to objective laws and practical requirements. Given that the proposed metric in this study is based on an existing one (Zheng and Huang, 2023b) whose qualitative analysis has been well reported, this section focuses on the analysis of the relationship between changes in flooding damage costs and flood resilience. Based on the analysis, we aim to determine whether the metric can satisfy the following requirements:

(1) Flood resilience based on the system performance curve should be dimensionless and nonnegative but not greater than 1.
(2) For two meshes with the same R_0 in Eq. (9), the mesh with higher flooding damage costs should exhibit lower flood resilience, representing its higher difficulty in reversing to the original economic state.
(3) If the flooding damage costs of a mesh equals 0, its economic state is completely unaffected by flooding and its flood resilience should be the same as that without considering flooding damage costs.
(4) The flood resilience of the mesh with the highest flooding damage costs in the study area should not be 0. This is because the given area gradually recovers to its original economic state due to the self-recovery and adaptive capacity of human society, although the recovery is slower compared to other areas.

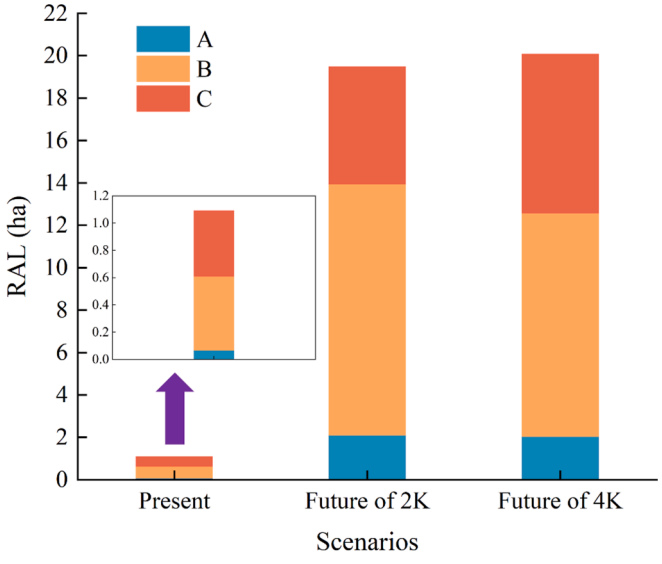


Fig.8. Urban flood resilience area loss (RAL) of different regions under present and future scenarios.

Based on the results, we demonstrate that the proposed metric satisfies the above requirements and can thus be used for flood resilience calculations and further analysis.

4.2. Urban flood resilience area loss

Fig. 8 presents the RAL of different regions. The total RAL in the study area increases with the temperature, with a dramatic increase from the present to future scenarios and a relatively limited increase from future scenarios 2 K to 4 K. This suggests that if no additional steps are taken to actively enhance flood resilience, the coverage of resilience losses in the study area will increase significantly in future periods

compared to the current scenario under the rainfall events of the same occurrence period. In addition, the increase of total RAL with temperatures when comparing the future scenarios indicates that continued climate change will further expand the coverage of flood resilience losses. This poses a greater challenge to the study area's capacity to address flooding issues.

In terms of specific regions, the RAL in all sub-regions under future scenarios increases markedly when compared with that under the present scenario. This indicates that climate change contributes to the expansion of the coverage of flood resilience loss on a global scale. The RAL in Regions A and B under the 4 K scenario is less than that under the 2 K scenario, while the opposite is observed in Region C. Further considering the maximum RAL in the study area under the scenario with the highest temperature, climate change contributes most significantly to the expansion of flood resilience losses in Region C. This is not conducive to the centralization of flood resilience enhancement measures in Region C. RAL in Region B is consistently the highest, demonstrating that targeted and effective measures in this region should be first taken when reducing the resilience area loss in the study area. In addition, considering the increasing resilience area loss in Region C with the development of climate change, the degree of change in flood resilience and the coverage of resilience loss in Region C should be given priority in flood prevention for long-term development.

4.3. Impacts of flooding damage costs on resilience

This study focuses on the flooding damage costs in areas with buildings. Thus, Fig. 9 depicts the impacts of flooding damage costs based on comparisons between the flood resilience results with and without considering damage costs of the overall study area and buildings. R and R_0 in Fig. 9 represent flood resilience with and without damage costs, which are calculated using the proposed metric and the existing one (Zheng and Huang, 2023b), respectively.

The flood resilience of the overall study area and buildings is reduced after considering the adverse effects of economic losses on the recovery capacity (Fig. 9). For the overall study area and the buildings, the difference between flood resilience with and without considering economic losses is small in the present scenario. However, there is a significant difference in flood resilience results with and without considering economic losses in future scenarios, particularly for buildings. This highlights the need to account for the impact of damage costs on flood resilience in light of climate change, as damage costs will have a non-negligible influence on the ability of an urban system to address flooding issues in the future. The differences in flood resilience of the overall study areas and buildings with and without economic losses are greatest for the 2 K scenario, followed by the 4 K scenario. This suggests that the negative impacts of economic losses on flood resilience will be more noticeable during rainfall events with higher rainfall intensities.

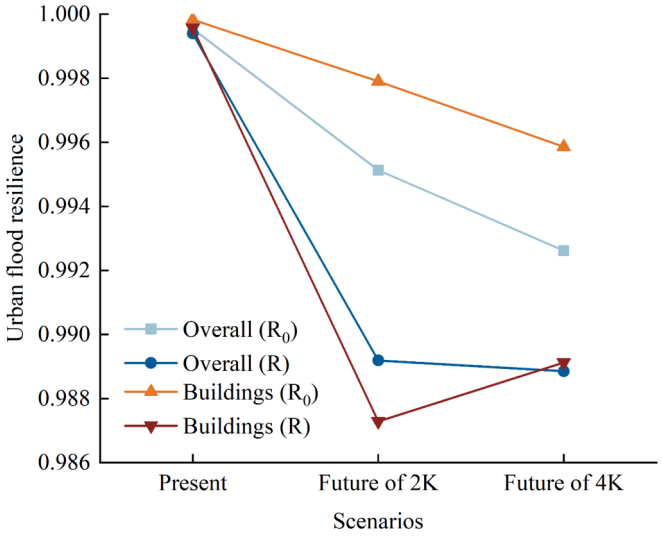


Fig.9. Urban flood resilience of the overall study area and buildings under present and future scenarios.

In terms of the overall study area, flood resilience is consistently highest in the present scenario and lowest in the 4 K scenario, regardless of whether damage costs are considered or not. After including damage costs, the difference between the 2 K and 4 K future scenarios becomes smaller. This can be attributed to the wider coverage of damage costs under the rainfall event of the 2 K scenario due to its higher peak rainfall intensity compared to the 4 K scenario (Fig. 5), despite the greater extent of damage costs under the 4 K scenario. When economic losses are not accounted for, the flood resilience of buildings is significantly higher than the overall flood resilience of the study area in all scenarios. This indicates that the functional characteristics of other land use types are more significantly affected in this case compared to buildings. However, when damage costs are considered, the flood resilience of buildings is lower than that of the overall study area under the 2 K scenario. The flood resilience of buildings under the 2 K scenario is also lower than that under the 4 K scenario. This suggests that in terms of the negative impacts on the resilience of buildings at high rainfall intensities, an increase in the area covered by damage costs is more significant compared to the greater damage costs in localized areas.

4.4. Suggestions for future research

Although this study considered the flooding socioeconomic impacts on urban flood resilience by including flooding damage costs in the performance-based metric, limitations still exist when applying the proposed metric. Constrained by the data availability and calculation complexity, flooding damage costs belong to the first-tier economic impacts based on official documents and similar research, neglecting the indirect costs. Indirect economic losses are described as the decrease in output that occurs both inside and outside of impacted regions due to shortages in supply, as well as modifications in demand and pricing signals brought about by the spread to other industries within the supply and trade network (Willner et al., 2018). Such indirect costs represent important flooding economic losses, yet they are difficult to measure (Newman and Noy, 2023). Future studies can therefore further consider the indirect economic losses when applying the proposed metric, with the aid of common assessment methods such as the input–output (IO) (Wang et al., 2017) and computable general equilibrium (CGE) (Tanoue et al., 2020) models, to make the flood resilience evaluation results more comprehensive and accurate.

5. Conclusions

A novel and rational performance-based flood resilience metric considering the flooding impacts on the urban economy was proposed and piloted in a typically highly urbanized area in Osaka City, Japan. The impacts of climate change on the flooding damage costs factor, system performance, and urban flood resilience were investigated based on the results from the 1D–2D hydrodynamic flood model. The flood resilience area loss and impacts of flooding damage costs on resilience were further discussed. The key findings are summarized as follows:

(1) Climate change exerts non-negligible pressure on the study area to maintain its economic characteristics during flooding. The higher the temperature, the lower the recovery and adaptive capacity of different types of regions.
(2) Rainfall intensity primarily affects the system's most unfavorable state, while the total amount and duration of rainfall mostly impact the system's relatively stable state following flooding. Similar trends can be observed between the system performance curves and rainfall event patterns.

(3) Climate change contributes to the shift in flood resilience from the highest class to lower classes and such shifts are more likely to occur between high and very high classes. The high peak rainfall intensity promotes the expansion of low-flood resilience areas.
(4) Climate change severity is positively correlated with the reduction in overall flood resilience. Continued climate change will further expand the coverage of flood resilience losses on a global scale.
(5) Damage costs will have a non-negligible influence on the ability of an urban system to address flooding issues in the future. The negative impacts of economic losses on flood resilience will be more noticeable during rainfall events with higher rainfall intensities. In terms of the negative impacts on the resilience of buildings at high rainfall intensities, an increase in the area covered by damage costs is more significant compared to the greater damage costs in localized areas.

Future research can further include indirect economic losses when applying the proposed metric to obtain more accurate flood resilience assessment results, supplementing the responses of flood resilience to climate change based on this study.

CRediT authorship contribution statement

Jiaxuan Zheng: Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Data curation, Conceptualization. **Xi Chen:** Visualization, Validation, Methodology, Investigation, Data curation. **Kenji Kawaike:** Visualization, Validation, Supervision, Software, Resources, Methodology. **Kazuki Yamanoi:** Visualization, Validation, Supervision, Methodology. **Takahiro Koshiba:** Supervision, Methodology. **Guoru Huang:** Writing – review & editing, Supervision, Resources, Project administration, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

This work was supported by the National Natural Science Foundation of China [grant number 52279015] and the State Key Laboratory of Subtropical Building and Urban Science [grant number 2023ZA01].

Data availability

Data will be made available on request.

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